

Statistical and Machine Learning

Linear models

Instructions

All questions should first be discussed in pairs or small groups. Afterwards, we shall discuss them together. Solutions will be distributed after each session.

Question 1 – Intro to Linear Models

Suppose that we observe the response Y_i , $i = 1, \dots, n$ with $n = 9$, along with numeric variables w_i and z_i , and a factor variable g_i that contains labels partitioning Y_i into three groups. For convenience, let $g_1 = g_2 = g_3 = 1$, $g_4 = g_5 = g_6 = 2$ and $g_7 = g_8 = g_9 = 3$ (we can always arrange the index so that this holds). We further assume that the following model is considered to be appropriate for the data:

$$Y_i = \beta_0 + \mu_{g_i} + \beta_1 w_i + \beta_2 z_i + \beta_3 z_i^2 + \beta_4 w_i z_i + \varepsilon_i, \quad i = 1, \dots, n.$$

Then, we may re-write the model as $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta}_0 + \boldsymbol{\varepsilon}$ by writing explicitly each element of $\mathbf{Y}$, $\mathbf{X}$, $\boldsymbol{\beta}_0$, $\boldsymbol{\varepsilon}$ by following these rules:

- Each factor level / group is assigned an indicator variable corresponding to a column in $\mathbf{X}$, such that:
 - $x_{ij} = 1$ for $j = 2, 3, 4$ indicates membership of the i -th observation to the $(j - 1)$ -th group
 - $x_{ij} \in \{0, 1\}$ and $\sum_{j=2}^4 x_{ij} = 1$
- Variables w_i and z_i are included in the model non-linearly.

a) Fill in the first three and the last element of the model in the matrix form below.

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta}_0 + \boldsymbol{\varepsilon},$$
$$\begin{bmatrix} Y_1 \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ Y_9 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 & 0 & w_1 & z_1 & z_1^2 & w_1 z_1 \\ 1 & 1 & 0 & 0 & w_2 & z_2 & z_2^2 & w_2 z_2 \\ 1 & 1 & 0 & 0 & w_3 & z_3 & z_3^2 & w_3 z_3 \\ 1 & 0 & 1 & 0 & w_4 & z_4 & z_4^2 & w_4 z_4 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ 1 & 0 & 0 & 1 & w_9 & z_9 & z_9^2 & w_9 z_9 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \mu_1 \\ \mu_2 \\ \mu_3 \\ \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \varepsilon_4 \\ \cdot \\ \cdot \\ \cdot \\ \varepsilon_9 \end{bmatrix}.$$

b) Answer the following questions:

- Is the model linear in the parameters? How can you conclude that?

Yes, parameters not squared, transformed, only a linear relationship with y .

- Is the model linear in errors? How can you conclude that?

Yes, only linear relationship with y .

- Is the model linear in regressors/features? How can you conclude that?

No, we can see that some features are squared or there is an interaction between them.

- How many parameters do we need to estimate?

At the first glance – we have 8 parameters. But actually, to fit this model, we need to drop one of the columns of $\mathbf{X}$ corresponding to one of the group variables (for example, by setting $\mu_1 = 0$ and removing the second column). Otherwise $\mathbf{X}$ would not be full rank.

Question 2 – Properties of the OLS estimator

Besides having a nice geometrical interpretation, the least squares estimator of β_0 has an *analytical expression with respect to $\mathbf{Y}$ and $\mathbf{X}$* . We can re-write the residual sum of squares as

$$S(\beta) = \|\mathbf{Y} - \mathbf{X}\beta\|^2 = (\mathbf{Y} - \mathbf{X}\beta)^\top (\mathbf{Y} - \mathbf{X}\beta)$$

a) Compute the minimizer $\hat{\beta}$ of function $S(\beta)$ and show that it is given by $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$.

Hint: use in your derivations the following properties

$$\frac{\partial}{\partial \mathbf{x}} \mathbf{x}^\top \mathbf{a} = \mathbf{a} \quad \text{and} \quad \frac{\partial}{\partial \mathbf{x}} \mathbf{x}^\top \mathbf{A} \mathbf{x} = 2\mathbf{A} \mathbf{x}.$$

$$\begin{aligned} S(\beta) &= \|\mathbf{Y} - \mathbf{X}\beta\|^2 = (\mathbf{Y} - \mathbf{X}\beta)^\top (\mathbf{Y} - \mathbf{X}\beta) \\ &= \mathbf{Y}^\top \mathbf{Y} - \mathbf{Y}^\top \mathbf{X}\beta - \beta^\top \mathbf{X}^\top \mathbf{Y} + \beta^\top \mathbf{X}^\top \mathbf{X}\beta. \end{aligned}$$

Note that $\mathbf{Y}^\top \mathbf{X}\beta$ and $\beta^\top \mathbf{X}^\top \mathbf{Y}$ are scalars. Since the transpose of a scalar is also a scalar, we have $\mathbf{Y}^\top \mathbf{X}\beta = \beta^\top \mathbf{X}^\top \mathbf{Y}$ and therefore

$$S(\beta) = \mathbf{Y}^\top \mathbf{Y} - 2\beta^\top \mathbf{X}^\top \mathbf{Y} + \beta^\top \mathbf{X}^\top \mathbf{X}\beta.$$

$S(\beta)$ is a *convex* function in β , and therefore the least squares estimator minimising the residual sum of squares, say $\hat{\beta}$, is obtained as a solution to the following equation:

$$\frac{\partial}{\partial \beta} S(\hat{\beta}) = -2\mathbf{X}^\top \mathbf{Y} + 2\mathbf{X}^\top \mathbf{X}\hat{\beta} = 0,$$

which follows from

$$\frac{\partial}{\partial \mathbf{x}} \mathbf{x}^\top \mathbf{a} = \mathbf{a} \quad \text{and} \quad \frac{\partial}{\partial \mathbf{x}} \mathbf{x}^\top \mathbf{A} \mathbf{x} = 2\mathbf{A} \mathbf{x}.$$

Solving the above yields

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.$$

b) Assuming that $\varepsilon \sim (\mathbf{0}, \sigma^2 \mathbf{I}_n)$, which yields $\mathbf{Y} \sim (\mathbf{X}\beta_0, \sigma^2 \mathbf{I}_n)$, prove that

c) $\hat{\beta}$ is unbiased, i.e., $\mathbb{E}(\hat{\beta}) = \beta_0$,

$$\mathbb{E}(\hat{\beta}) = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbb{E}(\mathbf{Y}) = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X}\beta_0 = \beta_0$$

.

ii) covariance of $\hat{\beta}$ is given by $\text{cov}(\hat{\beta}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$

$$\text{cov}(\hat{\beta}) = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \text{cov}(\mathbf{Y}) \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}.$$

- iii) Now, additionally, assume that $\boldsymbol{\varepsilon} \sim \mathcal{N}_n(\mathbf{0}, \sigma^2 \mathbf{I}_n)$ which yields $\mathbf{Y} \sim \mathcal{N}_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I}_n)$. Given that, why can you conclude that $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$ is also normally distributed?

Since $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$ is a linear transformation of $\mathbf{Y}$, it also follows a normal distribution:

$$\hat{\boldsymbol{\beta}} \sim \mathcal{N}_p(\boldsymbol{\beta}_0, \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}).$$

Question 3 – Properties of the Hat Matrix $\mathbf{H}$

Let $\mathbf{H} := \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ be the hat matrix defined in the course notes. Show that:

- (a) the trace of $\mathbf{H}$ is the number of identifiable parameters in the model, $\text{tr}(\mathbf{H}) = p$.

Hint: for any real matrix $\mathbf{A}$ it is always true that $\text{tr}(\mathbf{A}^\top \mathbf{A}) = \text{tr}(\mathbf{A} \mathbf{A}^\top)$.

$$\text{tr}(\mathbf{H}) = \text{tr}(\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top) = \text{tr}(\mathbf{X}^\top \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1}) = \text{tr}(\mathbf{I}_p) = p$$

- (b) $\mathbf{H}$ is *idempotent*: $\mathbf{H}\mathbf{H} = \mathbf{H}$,

$$\mathbf{H}\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top = \mathbf{H}$$

- (c) $\mathbf{H}$ is a *projection* matrix onto the space spanned by the columns of $\mathbf{X}$, and hence $\mathbf{H}\mathbf{X} = \mathbf{X}$,

$$\mathbf{H}\mathbf{X} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} = \mathbf{X}$$

Question 4 – Properties of Raw Residuals

Let $\mathbf{Y}$ a response vector, $\hat{\boldsymbol{\mu}} = \mathbf{X}\hat{\boldsymbol{\beta}}$ and let $\mathbf{H}$ be the hat matrix from Question 2 above.

- (a) define raw residuals in terms of $\mathbf{Y}$ and $\mathbf{H}$.

$$\mathbf{e} = \mathbf{Y} - \hat{\boldsymbol{\mu}} = \mathbf{Y} - \mathbf{H}\mathbf{Y} = (\mathbf{I}_n - \mathbf{H})\mathbf{Y}$$

- (b) compute $\mathbb{E}(\mathbf{e})$

$$\mathbb{E}(\mathbf{e}) = (\mathbf{I}_n - \mathbf{H})\mathbb{E}(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}_0 - \mathbf{H}\mathbf{X}\boldsymbol{\beta}_0 = \mathbf{X}\boldsymbol{\beta}_0 - \mathbf{X}\boldsymbol{\beta}_0 = \mathbf{0}$$

- (c) compute $\text{cov}(\mathbf{e})$

$$\text{cov}(\mathbf{e}) = \sigma^2 (\mathbf{I}_n - \mathbf{H})(\mathbf{I}_n - \mathbf{H})^\top = \sigma^2 (\mathbf{I}_n - \mathbf{H})$$

using that $\mathbf{H}$ is a projection matrix onto the column space of $\mathbf{X}$, as well as its idempotence and symmetry.

- (d) Assume $\mathbf{Y} \sim \mathcal{N}_n(\mathbf{X}\boldsymbol{\beta}_0, \sigma^2 \mathbf{I}_n)$. What is the distribution of $\mathbf{e}$? Why is so?

Raw residuals are linear transformations of $\mathbf{Y}$.

$$\mathbf{e} \sim \mathcal{N}_n(\mathbf{0}, \sigma^2 (\mathbf{I}_n - \mathbf{H})).$$

The normal distribution is degenerate, since $\mathbf{I}_n - \mathbf{H}$ is not of full rank. If $\mathbf{a} = \mathbf{H}\mathbf{c}$ for some $\mathbf{c} \in \mathbb{R}^n$, we have

$$\mathbf{a}^\top (\mathbf{I}_n - \mathbf{H})\mathbf{a} = 0,$$

which implies that some eigenvalues of $\mathbf{I}_n - \mathbf{H}$ are zero.

Question 5 – Influential Observations

- (a) What is the difference between an **outlier** and an **influential observation** in a regression context?

Model answer: An outlier is an observation with an unusually large residual — its y value is far from the fitted value $\hat{y}_i$. An influential observation, by contrast, is one with an unusual x_i value (high leverage) such that removing it substantially changes the fitted model. A point can be an outlier without being influential (if it has low leverage), or influential without being a traditional outlier (if it lies far from the other x values but is consistent with the fitted line).

- (b) What does the leverage h_{ii} measure, and what is the commonly used threshold for declaring an observation as high leverage?

Model answer: The leverage h_{ii} is the i -th diagonal element of the hat matrix $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$. It measures how much the fitted value $\hat{\mu}_i$ is “pulled” toward the observed y_i : as $h_{ii} \rightarrow 1$, $\text{Var}(e_i) \rightarrow 0$, so the model is forced to fit that point exactly. The threshold commonly used is $h_{ii} > 2p/n$, i.e., twice the average leverage.

- (c) Cook’s distance D_i combines two sources of information. Name them and explain why both are necessary.

Model answer: Cook’s distance defined as

$$D_\ell = \frac{\sum_{i=1}^n (\hat{\mu}_i^{(\ell)} - \hat{\mu}_i)^2}{p \hat{\sigma}^2} = s_\ell^2 \frac{h_{\ell\ell}}{(1 - h_{\ell\ell})p}$$

combines (i) the **standardized residual size** s_i (how far the observation is from the fitted model) and (ii) the **leverage** h_{ii} (how unusual the x value is). Both are needed: a large residual with low leverage will not move the regression line much; a high-leverage point with a small residual sits near the fitted line and therefore also has little effect. Only the combination of both — high leverage *and* large residual — results in a truly influential observation.

- (d) A practitioner finds Cook’s distance $D_i = 0.8$ for a single data point. What action, if any, would you recommend?

Model answer: $D_i = 0.8 > 0.5$ flags this point as **potentially influential**. The recommended might be: (1) Inspect the observation for data entry errors or other anomalies; (2) Re-fit the model with the observation removed and compare the parameter estimates; (3) If estimates change substantially, consider whether the point is legitimately part of the population of interest or whether it should be removed / modelled separately. Since $D_i < 1$, it is not yet classified as *highly* influential, so removal should not be automatic.

Question 6 — OLS as Best Linear Approximation

- (a) Write down the population-level minimisation problem that defines the OLS parameter β_0 , without assuming the true model is linear.

Model answer:

$$\beta_0 = \arg \min_{\beta} \mathbb{E}[(Y - \mathbf{X}^\top \beta)^2]$$

This is well-defined purely from the joint distribution of $(Y, \mathbf{X})$ — no assumption about the functional form of $f(\mathbf{X}) = \mathbb{E}(Y | \mathbf{X})$ is needed.

- (b) Suppose the true relationship is $Y = f(\mathbf{X}) + \varepsilon$. Write Y in the form $Y = \mathbf{X}^\top \beta + r(\mathbf{X}) + \varepsilon$. What does $r(\mathbf{X})$ represent, and what orthogonality condition must it satisfy?

Model answer: $r(\mathbf{X}) = f(\mathbf{X}) - \mathbf{X}^\top \beta$ is the **approximation error** — the part of the conditional mean $f(\mathbf{X})$ not captured by the linear predictor. Under exogeneity $\mathbb{E}(\varepsilon | X) = 0$, it must satisfy:

$$\mathbb{E}\{\mathbf{X} r(\mathbf{X})\} = 0$$

This says $r(\mathbf{X})$ is orthogonal to every predictor $\mathbf{X}$, so all the linear association between $\mathbf{X}$ and Y is captured by $\mathbf{X}^\top \beta$, and the remainder is genuinely nonlinear.

- (c) Why does the moment condition $\mathbb{E}\{X(Y - \mathbf{X}^\top \beta)\} = 0$ make OLS estimators meaningful even under misspecification?

Model answer: The condition defines β_0 through a **moment equation** rather than through a correctly specified likelihood. As long as this moment condition holds — which it does by construction under exogeneity — the OLS estimator $\hat{\beta}$ is consistent for the best linear predictor β_0 regardless of whether the true $f(\mathbf{X})$ is linear. This is the basis of M-estimation and GMM: we are estimating a parameter of the data-generating process, not necessarily the parameter of a correctly specified model.

- (d) Why is the statement “OLS only works under linearity” too strong?

Model answer: Even without a linear true model, $\beta_0 = \arg \min_{\beta} \mathbb{E}(Y - \mathbf{X}^\top \beta)^2$ is well-defined as the parameter of the **best linear approximation** to $\mathbb{E}(Y | \mathbf{X})$. The OLS estimator $\hat{\beta}$ is consistent for β_0 as long as the orthogonality/moment condition $\mathbb{E}\{\mathbf{X}(Y - \mathbf{X}^\top \beta_0)\} = 0$ holds — which it does under exogeneity, regardless of whether $f(\mathbf{X})$ is linear. The model is misspecified, but $\hat{\beta}$ still consistently estimates a meaningful, interpretable quantity.

Question 7 — Logistic Regression: Concepts

- (a) Give three reasons why a standard linear regression model is inappropriate for $Y \in \{0, 1\}$.

Model answer:

1. The linear predictor $\mathbf{x}^\top \beta$ can take any value in $\mathbb{R}$, including values outside $[0, 1]$, which are impossible for a probability.
2. The errors $\varepsilon_i = Y_i - \pi_i$ cannot be homoscedastic: for a Bernoulli variable, $\text{Var}(Y_i | \mathbf{x}_i) = \pi_i(1 - \pi_i)$, which depends on $\mathbf{x}_i$.
3. The linear model does not return a binary (or probability) quantity — it models the response directly as a continuous linear function, which is not appropriate for a 0/1 outcome.

- (b) Write down the logit link $g(\pi_i)$ and inverse logit $h(\eta)$. Describe what h does geometrically.

Model answer:

$$g(\pi_i) = \log \frac{\pi_i}{1 - \pi_i} = \mathbf{x}_i^\top \beta, \quad h(\eta) = \frac{e^\eta}{1 + e^\eta} = \frac{1}{1 + e^{-\eta}}$$

Geometrically, h is a **sigmoid (S-shaped) curve** that smoothly compresses the entire real line into the open interval $(0, 1)$. It is symmetric around $h(0) = 0.5$ and has horizontal asymptotes at 0 and 1.

- (c) Suppose $\hat{\beta}_1 = 1.6$. By how much do the odds and log-odds change?

Model answer: A one-unit increase in x_1 (holding other variables constant):

- **Odds** are multiplied by $e^{1.6} \approx 4.95$ — i.e., the odds increase by a factor of approximately 5.
- **Log-odds** increase by the additive constant 1.6.

- (d) State the score equation and its relation to cross-entropy.

Model answer: The score equation is:

$$s(\hat{\beta}_{\text{ML}}) = \sum_{i=1}^n (Y_i - h(\mathbf{x}_i^\top \hat{\beta}_{\text{ML}})) \mathbf{x}_i = \mathbf{0}$$

The cross-entropy loss is $\text{CE}(\beta) = -\log L(\beta; y)$. Maximising the log-likelihood is equivalent to minimising CE, so the score equation is also the first-order condition (gradient = 0) of the cross-entropy minimisation. The score measures the residuals $Y_i - \hat{\pi}_i$ weighted by $\mathbf{x}_i$, analogously to the OLS normal equations.

- (e) How to construct a 95% CI for β_k , and interpretation of 0.

Model answer: By the asymptotic normality of the MLE:

$$\hat{\beta}_k \pm z_{0.025} \cdot \widehat{\text{s.e.}}(\hat{\beta}_k)$$

where $z_{0.025} \approx 1.96$ and $\widehat{\text{s.e.}}(\hat{\beta}_k) = \sqrt{[\hat{\mathcal{I}}(\hat{\beta})^{-1}]_{kk}}$. If the interval contains 0, we **fail to reject** $H_0 : \beta_k = 0$ at the 5% level, meaning there is insufficient evidence that x_k has a linear effect on the log-odds of Y .

Question 8 — LDA: Concepts

- (a) Describe the generative approach of LDA.

Model answer: LDA models the distribution of X **within each class** using a class-conditional density $f_k(x) = f_{X|Y}(x | Y = k)$ and a prior $P(Y = k) = \pi_k$. Classification uses Bayes' rule to compute posterior probabilities:

$$P(Y = k | X = x) = \frac{\pi_k f_k(x)}{\sum_{\ell=1}^K \pi_\ell f_\ell(x)}$$

A new observation is assigned to the class with the highest posterior probability.

- (b) Why Gaussian assumption and shared covariance in two classes helps us to get a linear boundary.

Model answer: LDA assumes $X | Y = k \sim \mathcal{N}(\mu_k, \sigma^2)$ (same σ^2 for all k). When computing $\log p_1(x) - \log p_2(x)$, the quadratic terms $x^2/(2\sigma^2)$ cancel because the covariance is shared, leaving a decision rule that is **linear** in x . If each class had its own σ_k^2 (Quadratic Discriminant Analysis), the quadratic terms would not cancel and the boundary would be quadratic.

- (c) How can you understand the Fisher's LDA optimization criterion.

Model answer: Fisher's LDA chooses the projection vector $\mathbf{w}$ to maximise so called **Rayleigh quotient**:

$$\arg \max_{\mathbf{w}} \frac{S_B}{S_{\text{in}}} = \arg \max_{\mathbf{w}} \frac{\mathbf{w}^\top (\mu_1 - \mu_2)(\mu_1 - \mu_2)^\top \mathbf{w}}{\mathbf{w}^\top (\Sigma_1 + \Sigma_2) \mathbf{w}}$$

$S_B = (m_1 - m_2)^2$ is the **between-class scatter**: the squared distance between projected class means. $S_{\text{in}} = \sum_{z_i \in \text{class } 1} (z_i - m_1)^2 + \sum_{z_i \in \text{class } 2} (z_i - m_2)^2$ is the **within-class scatter**: the total spread of points around their own class mean in the projected space.

The intuition is as follows: we want a projection direction $\mathbf{w}$ such that the two class means are **far apart** (large S_B) while points within each class remain **tightly clustered** (small S_{in}). Maximising the ratio ensures both goals are pursued simultaneously — neither can be trivially satisfied by simply rescaling $\mathbf{w}$, since the quotient is scale-invariant.

When the class-conditional distributions are Gaussian with shared covariance Σ , the optimal solution is:

$$\mathbf{w} \propto \Sigma^{-1}(\mu_1 - \mu_2)$$

This is precisely the linear discriminant returned by `lda()` in R. The corresponding decision boundary is the hyperplane **perpendicular to $\mathbf{w}$** passing through the midpoint of the two class means.

Question 9 — LDA: R Output & Decision Boundary

Consider the output of the LDA on slide 59.

- (a) Using slide 60, compute the midpoint $\text{mid} = (\text{mid}_1, \text{mid}_2)$

Model answer:

$$\text{mid} = \frac{(4.260, 1.326) + (5.552, 2.026)}{2} = \left(\frac{9.812}{2}, \frac{3.352}{2} \right) = (4.906, 1.676)$$

- (b) Using slide 60, compute the equation of the decision boundary.

Model answer: With $\mathbf{w} = (w_1, w_2) = (0.871, 2.925)$ and $\text{mid} = (4.906, 1.676)$:

$$\text{slope} = -\frac{w_1}{w_2} = -\frac{0.871}{2.925} \approx -0.298$$

$$\text{intercept} = \text{mid}_2 - \text{slope} \times \text{mid}_1 = 1.676 - (-0.298)(4.906) \approx 1.676 + 1.462 = 3.138$$

$$\Rightarrow x_2 \approx -0.298 x_1 + 3.138$$

- (c) Why do both LDA and logistic regression produce linear decision boundaries?

Model answer:

- **Logistic regression:** The log-odds $\log[\pi/(1 - \pi)] = \mathbf{x}^\top \boldsymbol{\beta}$ is linear in x by construction. The decision boundary (where $P(Y = 1 | \mathbf{x}) = 0.5$) corresponds to $\mathbf{x}^\top \boldsymbol{\beta} = 0$, which is a hyperplane.
- **LDA:** The decision boundary arises from $\log p_1(\mathbf{x}) = \log p_2(\mathbf{x})$. Under the Gaussian assumption with shared covariance, all quadratic terms in x cancel, leaving a linear equation in x .

- (b) When does each method outperform the other?

Model answer: LDA outperforms logistic regression when the true class-conditional distributions are approximately Gaussian with equal covariance, especially in small samples where the stronger parametric assumption provides efficiency gains. *Example: classifying patients from blood test measurements that are approximately normally distributed.*

Logistic regression outperforms LDA when the normality assumption is violated (e.g., skewed, binary, or discrete predictors), since it only models the conditional probability $P(Y = 1 | \mathbf{x})$ directly without assuming a distribution for X . *Example: classifying fraud/non-fraud from transaction amounts, which are typically heavy-tailed.*